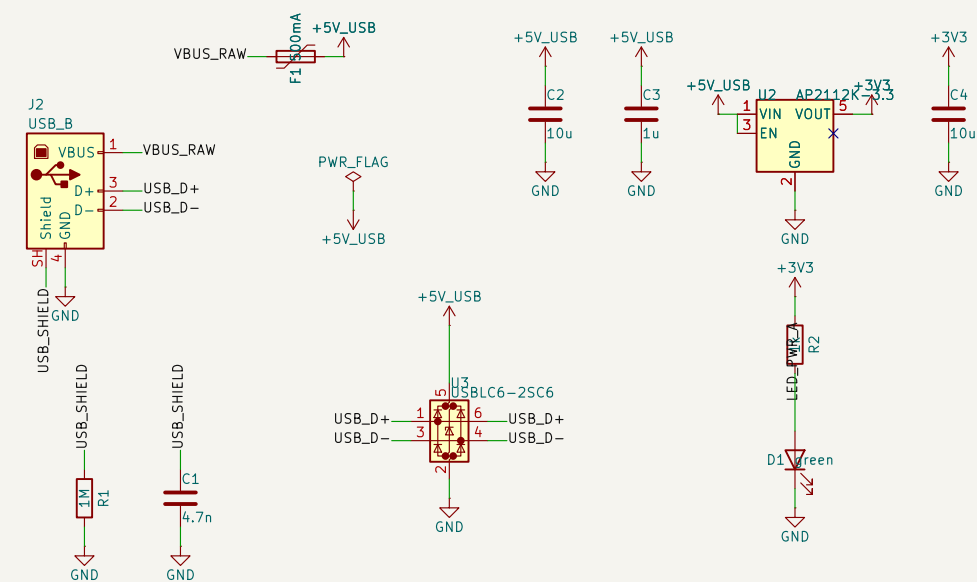


USB INPUT, PROTECTION, 3.3 V



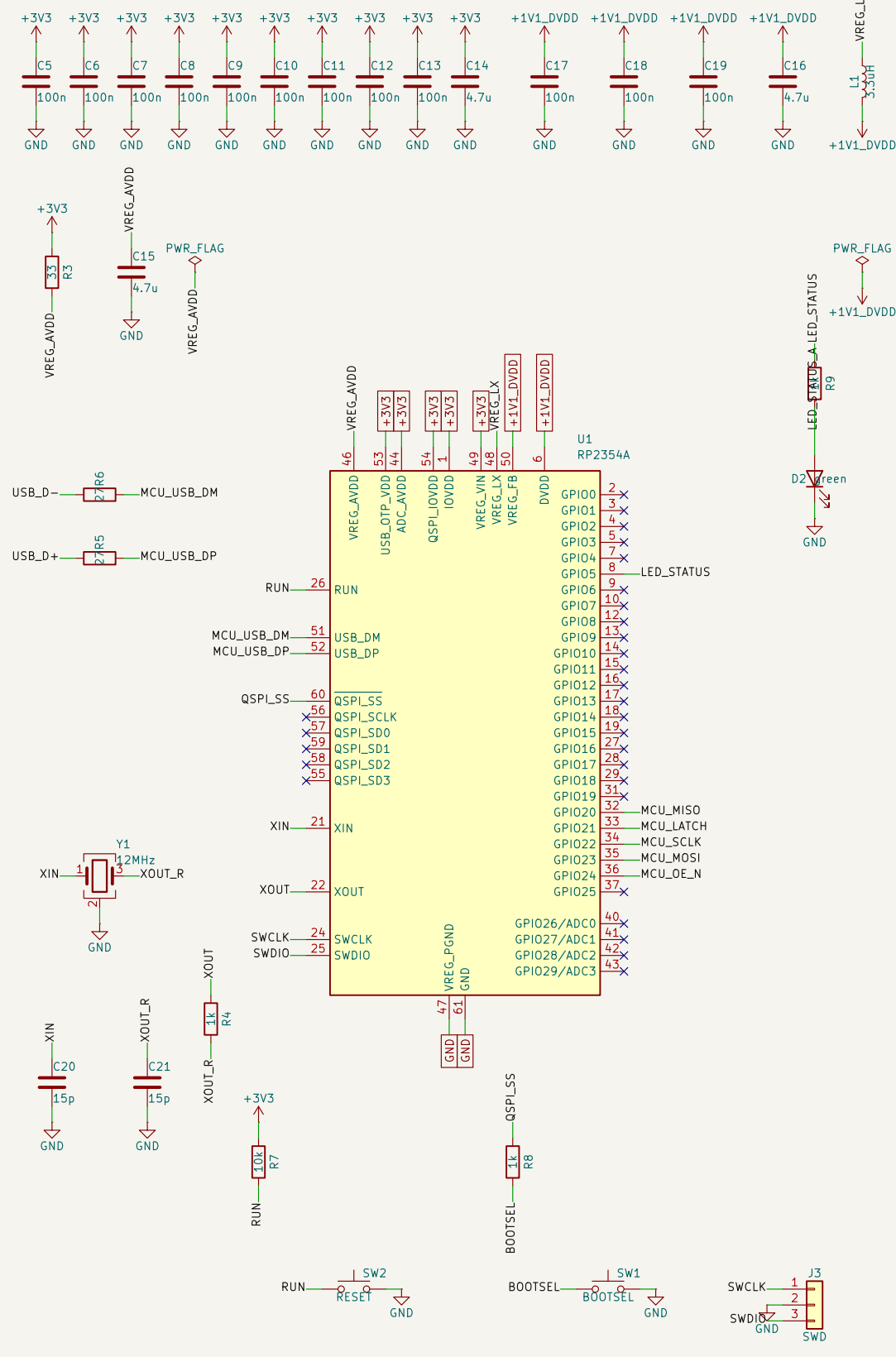
NOTES

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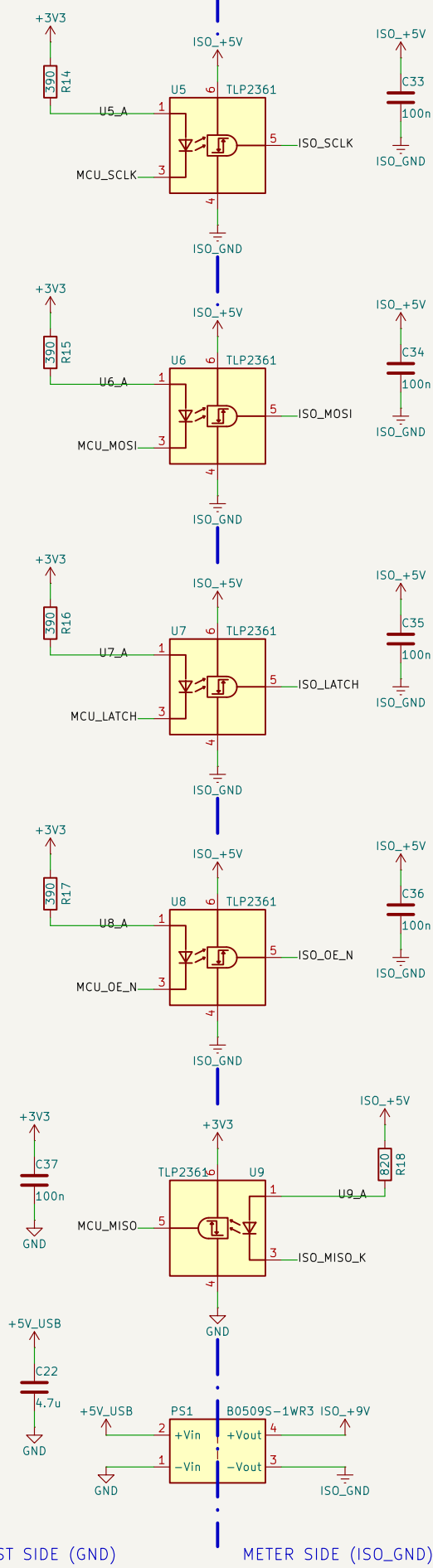
Isolated serial link (RP235A4 SPI0 / PIO0): SPI mode 0, MSB first, c = 1 MHz, 40 clocks per transfer.
Pulse LATCH low >= 1 us, then clock 5 bits. LATCH low: the 74HC1655 sample the meter outputs.
LATCH rising edge: the 74HC595 outputs take the last two bytes clocked in.
/OE (GPIO24) high = command outputs off (the default while the MCU is unpowered or in reset).

MISO byte0 u10: POL+ POL- LATCH_FUNC_B RANGE2 RANGE2_RANGE1 PRINT_PULSE
byte1 u11: 10'3 digit, 10'4 digit, 10'2 digit (8 4 2 1)
byte2 u12: 10'5 digit, 10'4 digit
byte3 u13: 10'1 digit, 10'0 digit
byte4 u15: PRINT_LEVEL, DATA_CAN_CHANGE, OVERLOAD 1x0'6 1 0 1 0 (link check)
MOSI byte0 u1: INT0, INT1, INT2, INT3, INT4, INT5, INT6, INT7, INT8, INT9, INT10, INT11, INT12, INT13, INT14, INT15, INT16, INT17, INT18, INT19, INT20, INT21, INT22, INT23, INT24, INT25, INT26, INT27, INT28, INT29, INT30, INT31, INT32, INT33, INT34, INT35, INT36, INT37, INT38, INT39, INT40, INT41, INT42, INT43, INT44, INT45, INT46, INT47, INT48, INT49, INT50, INT51, INT52, INT53, INT54, INT55, INT56, INT57, INT58, INT59, INT60, INT61, INT62, INT63, INT64, INT65, INT66, INT67, INT68, INT69, INT70, INT71, INT72, INT73, INT74, INT75, INT76, INT77, INT78, INT79, INT80, INT81, INT82, INT83, INT84, INT85, INT86, INT87, INT88, INT89, INT90, INT91, INT92, INT93, INT94, INT95, INT96, INT97, INT98, INT99, INT100, INT101, INT102, INT103, INT104, INT105, INT106, INT107, INT108, INT109, INT110, INT111, INT112, INT113, INT114, INT115, INT116, INT117, INT118, INT119, INT120, INT121, INT122, INT123, INT124, INT125, INT126, INT127, INT128, INT129, INT130, INT131, INT132, INT133, INT134, INT135, INT136, INT137, INT138, INT139, INT140, INT141, INT142, INT143, INT144, INT145, INT146, INT147, INT148, INT149, INT150, INT151, INT152, INT153, INT154, INT155, INT156, INT157, INT158, INT159, INT160, INT161, INT162, INT163, INT164, INT165, INT166, INT167, INT168, INT169, INT170, INT171, INT172, INT173, INT174, INT175, INT176, INT177, INT178, INT179, INT180, INT181, INT182, INT183, INT184, INT185, INT186, INT187, INT188, INT189, INT190, INT191, INT192, INT193, INT194, INT195, INT196, INT197, INT198, INT199, INT200, INT201, INT202, INT203, INT204, INT205, INT206, INT207, INT208, INT209, INT210, INT211, INT212, INT213, INT214, INT215, INT216, INT217, INT218, INT219, INT220, INT221, INT222, INT223, INT224, INT225, INT226, INT227, INT228, INT229, INT230, INT231, INT232, INT233, INT234, INT235, INT236, INT237, INT238, INT239, INT240, INT241, INT242, INT243, INT244, INT245, INT246, INT247, INT248, INT249, INT250, INT251, INT252, INT253, INT254, INT255, INT256, INT257, INT258, INT259, INT260, INT261, 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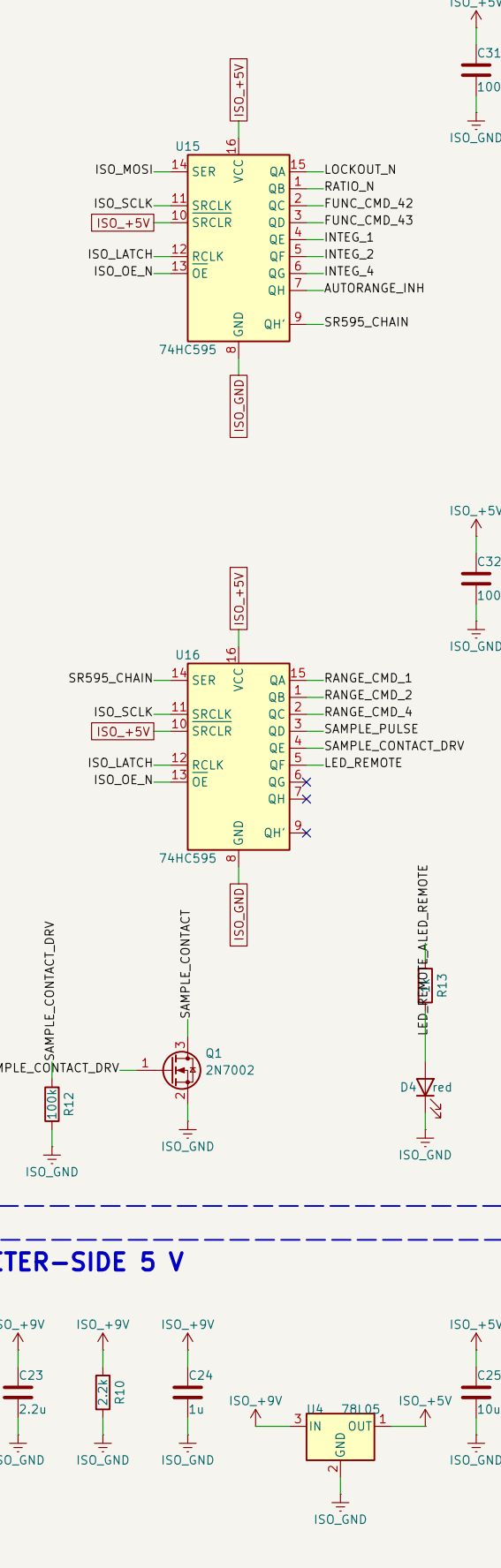
RP2354A (2 MB IN-PACKAGE FLASH)



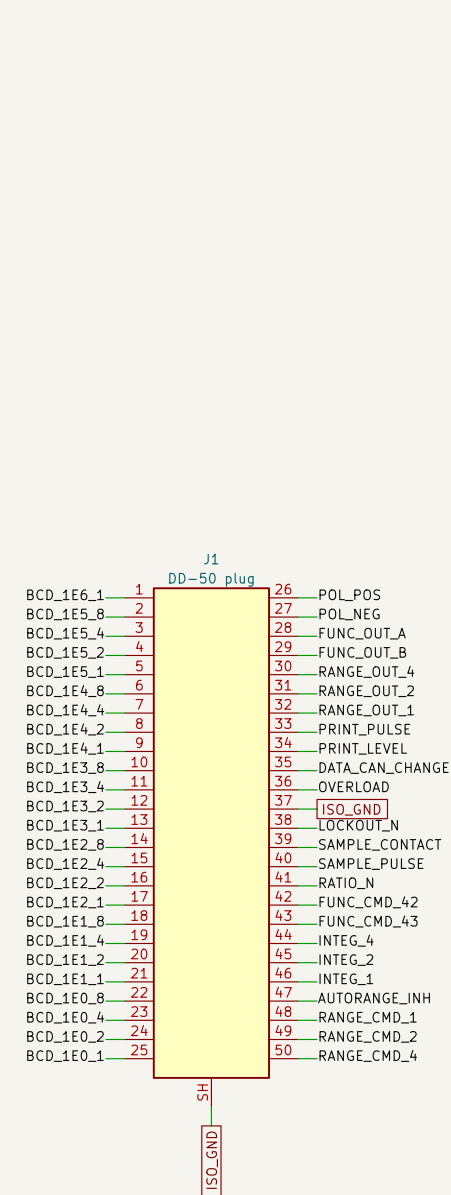
ISOLATION: 5 x TLP2361 + B0509S



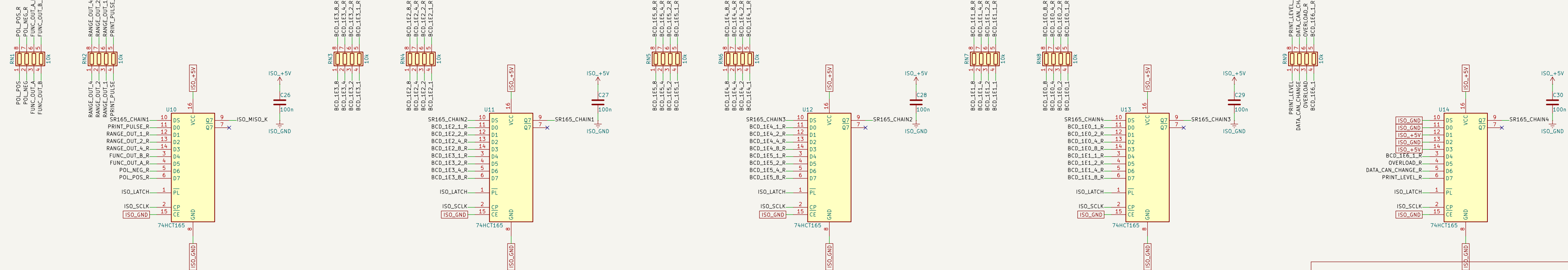
COMMAND OUTPUTS: 2 x 74HC595



J1: DD-50 PLUG TO 7075 SKB



DISPLAY DATA / STATUS INPUTS: 5 x 74HCT165 (MISO = U10.Q7; chain U10 <- U11 <- U12 <- U13 <- U14)



Net labels with the same name are connected
Generated from the SKiDL netlist by scripts/gen_schematic.py

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File: solartron_7075_interface.kicad_sch		
Title: Solartron 7075 USB interface (SKiDL)		
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KiCad E.D.A. 10.0.6		Id: 1/1